

Universal phenomenology, divergent mechanism: a behavioural gradient of mobility from foraging to commute

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Abstract

Three influential scaling laws of human mobility — the truncated power law of step lengths and the bounded radius of gyration¹, the universal visitation law $\rho \propto (rf)^{-2}$ ², and the hierarchical-container architecture³ — were each established on datasets of 10^5 – 10^7 humans and have since been treated as a human-specific phenomenology. Animal-movement ecology has developed in parallel around a different paradigm based on Lévy-flight foraging^{4–6}, optimal foraging theory^{7,8} and home-range allometry⁹, with persistent statistical controversies^{10,11} and few attempts at a unified formulation. Here we apply a single estimator battery to GPS records of 14 African elephants, 25 Northern gannets and 92 white storks tracked for up to a decade, and compare the results with the published human values. We find that the phenomenological architecture is universal: all three species exhibit truncated power-law step-length distributions, logarithmic saturation of $\langle r_g(t) \rangle$, hierarchical containers with four to five lognormal scales, and circadian peaks of return. The underlying microscopic mechanism, however, is not. The Song preferential-return scaling relations¹² hold closely only when the visitation Zipf exponent ζ approaches unity, defining a behavioural gradient $\zeta = 0.28$ (elephant) $\rightarrow 0.86$ – 0.89 (stork, gannet) $\rightarrow 1.0$ (human). The Schlöpfer visitation law with exponent $\eta = 2$ ² is a regime rather than a universal: species at near-constant locomotor speed return $\eta \approx 1$, whereas humans on motorised infrastructure return $\eta \approx 2$. Restricting the stork record to the obligate central-place breeding window yields a tail exponent $\alpha = 1.70 \pm 0.05$ that coincides with the human value 1.75 ± 0.15 ¹. Humans are therefore not a separate class of mobile organism but the extremum of a continuous behavioural-energetic axis — a result that unifies Christaller’s central-place theory¹³ and Charnov’s marginal-value theorem⁸ under a single physical principle.

Companion repository. All figure-level numerical data (CSV), the compiled manuscript, the cover letter, the preregistration and a master table of fitted exponents with 95% bootstrap confidence intervals are archived at <https://github.com/david888azv/universal-mobility-gradient>. The repository is organised by species: per-figure CSV files are placed in `data/elephant/`, `data/gannet/` and `data/stork/` under the naming pattern `fig<N>_<quantity>_<species>.csv`, so that the numerical content of any main-text panel can be located directly from its filename. Cross-species summary tables and the unified-model sweep sit in `data/cross_species/`; per-species intermediate diagnostics from each of the four estimator blocks (González, Song, Alessandretti, Schlöpfer) are in `figures/supplementary/<species>/`. The analysis scripts are available from the corresponding author upon request.

Introduction

Over the past two decades the physics of human mobility has converged on a compact set of empirical laws derived from massive longitudinal datasets. Pioneering work on dollar-bill diffusion¹⁴ and mobile-phone traces¹ established that the step-length distribution $P(\Delta r)$ is a truncated power law with exponent $\alpha \approx 1.75$, that the radius of gyration $r_g(t)$ saturates logarithmically rather than diffusing as $t^{1/2}$, and that individuals return to a small set of preferred locations whose visit frequencies follow Zipf’s law $P(L) \sim 1/L$. A microscopic generative model — exploration at decreasing rate $P_{\text{new}} = \rho S^{-\gamma}$ coupled with preferential return $\Pi_i = f_i$ — predicts three independent scaling relations among the empirical exponents¹². More recent analyses have decomposed mobility into a hierarchy of nested spatial containers³, and aggregate flows have been shown to obey a universal visitation law $\rho(r, f) = \mu/(rf)^2$ derivable from a constant collective effort budget $\frac{1}{2}\rho v^2 = \text{const}$ ². All four laws are influential, but they have been validated almost exclusively on humans in cities.

Animal-movement ecology has in parallel built a distinct paradigm. Beginning with the report of Lévy-flight foraging in wandering albatrosses⁴, the framework was extended to bumblebees and deer⁴, spider monkeys¹⁵, marine predators^{5,6} and other taxa, under the hypothesis that Lévy flights optimise random search for sparsely distributed prey¹⁶. The framework was sharply contested by Edwards et al.¹⁰, who showed that the original albatross, bumblebee and deer evidence collapsed under maximum-likelihood power-law fitting, and the field has since debated which species genuinely exhibit Lévy behaviour, and when^{11,17}. The unifying movement-ecology paradigm of Nathan et al.¹⁸ emphasised internal state, motion capacity and external drivers without committing to any particular scaling law, and large-scale tracking infrastructure has matured around the Movebank repository^{19,20}, opening the possibility of cross-species statistical synthesis.

Despite parallel transitions to sensor-rich data, there has been little cross-fertilisation between human-mobility physics and animal-movement ecology²¹. Hu et al.²² proposed a general optimisation framework in which home-return constraints generate exponents between 1 and 2 but did not test it empirically across taxa. Yan et al.²³ developed a universal mobility model for diverse spatial scales, though the empirical comparison remained within human datasets. Tikhonov & Strijkers²⁴ tested human scaling laws on 11 European Herring Gulls and reported a striking similarity between bird and human visitation distributions during the non-breeding season, with strong divergence during breeding. Their case study is the closest antecedent to the present work but is limited to a single species and a subset of the laws.

Here we extend this line of work along three axes. First, we apply a unified estimator battery covering the four human-mobility laws of references^{1–3,12} to three GPS-tracked vertebrate species spanning a behavioural gradient, from a home-ranging non-central-place forager (the African elephant *Loxodonta africana*) through a colonial central-place breeder (the Northern gannet *Morus bassanus*) to an obligate central-place migrant (the white stork *Ciconia ciconia*), and we compare the results with published human reference values. Second, we report bootstrap confidence intervals on every metric to assess robustness. Third, we propose a unified Song–Alessandretti–Schläpfer model with two parameters — central-place strength γ_{CP} and velocity heterogeneity η_v — that places all four species on a continuous parameter manifold. The central

claim is that humans are not a different class of mobile organism but an extremum of a continuous behavioural-energetic axis: the mobility laws appear universal where they are measured — in their phenomenology — and split into a clean gradient where they are mechanistic.

Results

Step-length distributions: common form, taxon-specific exponent

We computed the pooled step-length distribution $P(\Delta r)$ for each species after equirectangular projection to local Cartesian coordinates, and discarded consecutive identical positions. Maximum-likelihood power-law fits following Clauset, Shalizi & Newman²⁵ yield Kolmogorov–Smirnov (KS) distances below 0.07 for all three species. The fitted tail exponents are $\alpha_{\text{elephant}} = 3.40 \pm 0.18$, $\alpha_{\text{gannet}} = 2.83 \pm 0.32$, $\alpha_{\text{stork}} = 3.48 \pm 0.12$ (95% bootstrap CI by individual; Fig. 1a–c). The published human value is $\alpha_{\text{human}} = 1.75 \pm 0.15$ ¹. Animal exponents are uniformly steeper than human, consistent with stronger localisation within home ranges (Fig. 1d); the rank order tracks the magnitude of the home range relative to a single central place. The lower cut-offs $x_{\text{min}} = 1, 9$ and 3 km respectively are consistent with the expected home-range diameters of the three species.

A stringent robustness test is the seasonal split of the stork record. White storks alternate between a breeding phase (April–August, latitude greater than 40° N, chicks tied to the nest) and a migratory/wintering phase (latitudes 11°–48° N). Pooling all phases yields $\alpha = 3.48$; restricting the same estimator to the breeding phase alone yields $\alpha_{\text{breed}} = 1.70 \pm 0.05$, statistically indistinguishable from the human value 1.75 ± 0.15 (Fig. 4a). When the bird is constrained to a single central place by reproductive obligation, its step-length distribution converges onto the urban-human pattern. This supports the regime interpretation: phenomenology converges with central-place strength rather than with taxonomy.

A quantitative Zipf gradient ζ

The visitation rank–frequency relation $P(L) \sim L^{-\zeta}$ exhibits a pronounced gradient across species (Fig. 2a). Using a 500 m grid and top-50 pooled ranks across individuals (the convention of refs^{1,12}), we obtain $\zeta_{\text{elephant}} = 0.28 \pm 0.07$, $\zeta_{\text{gannet}} = 0.89 \pm 0.05$ and $\zeta_{\text{stork}} = 0.86 \pm 0.04$. The Hu–Yan home-return framework^{22,23} predicts $\zeta \rightarrow 1$ in the strong-CP limit, matching the human value $\zeta_{\text{human}} \approx 1.0$ ¹. The ordering is preserved at grids of 200 m and 1 km, and when the statistic is computed per individual (each animal’s own rank-1 site as anchor), storks rise to $\zeta = 1.01$, matching the human exactly.

The biological interpretation is direct. The elephant uses its home range relatively uniformly; no single watering hole or feeding site dominates. The gannet returns repeatedly to its nesting colony and to a small number of offshore foraging hotspots, producing strong but spatially distributed central-place behaviour. The stork’s chicks force it back to the nest several times per day, generating the most concentrated visitation pattern of any animal we measured. Humans, under the polarity of home and work, sit at the natural maximum of this axis.

The Song scaling relations: validity follows ζ

Song et al.¹² derived three predictions from the exploration-and-preferential-return (EPR) model: $\mu = \beta/(1 + \gamma)$, $\zeta = 1 + \gamma$ and $\beta = \mu\zeta$, where μ is the exponent of $S(t) \sim t^\mu$, β the waiting-time exponent and γ the exploration-decay exponent. We measured all four quantities per species via block-bootstrap by individual (Methods) and tested each relation as a discrepancy ratio $\max(\text{prediction}, \text{measurement})/\min(\text{prediction}, \text{measurement})$. For humans, the three relations hold with ratio close to unity¹². Across the animals, the median discrepancy ratio decreases monotonically as ζ rises: elephant $\sim 5.5\times$, gannet $\sim 1.3\times$ and stork $\sim 2.5\times$ (the latter elevated by mixing breeding and migration phases that violate the quasi-stationarity assumption of the EPR model; Fig. 3a). Restricting the stork record to the breeding phase alone reduces the worst ratio to $2.2\times$, comparable to the gannet, and confirms that the residual mismatch reflects regime mixing rather than mechanism failure.

We interpret this pattern as evidence that Song’s preferential-return mechanism $\Pi_i = f_i$ is not a universal feature of mobile organisms but emerges with central-place obligation. Elephant exploration appears resource-driven rather than memory-driven; the exploration-decay exponent γ is small and the trajectory is close to Brownian. Gannet nest fidelity introduces a meaningful preferential-return component, entering the regime where Song’s relations become approximate. Obligate central-place species — breeding storks and humans with home–work polarity — approach the original derivation.

The visitation law η : a transport-regime split

The Schl pfer visitation law $\rho(r, f) = \mu/(rf)^\eta$ with $\eta = 2$ is derived from a constant collective effort budget $\frac{1}{2}\rho v^2 = \text{const}$, where $v \equiv rf$ is the effective velocity². This argument applies to a regime in which transport cost scales kinetically with velocity — a feature of motorised infrastructure but not of biological locomotion at near-constant speed. We measured η directly on a 1 km grid with $T = 4$ -month observation windows following the Schl pfer protocol. Bootstrap 95% CI give $\eta_{\text{elephant}} = 1.03 \pm 0.04$, $\eta_{\text{gannet}} = 0.65 \pm 0.07$ and $\eta_{\text{stork}} = 1.09 \pm 0.05$, compared with $\eta_{\text{human}} = 2.05 \pm 0.02$ ² (Fig. 3b). The Schl pfer invariance test of mean effective distance per visitor against the number of visitors holds for all three animal species ($R^2 < 0.10$), consistent with the same mathematical form $\rho \propto (rf)^{-\eta}$ but with species-specific exponent.

The split is not taxonomic (mammal versus bird) nor by locomotor mode (walking versus flying), but by velocity heterogeneity. Elephants walk at approximately 5 km h^{-1} regardless of trip length; gannets glide at roughly 50 km h^{-1} regardless of trip length. Both produce $\eta \approx 1$ (linear-cost regime with $\rho v \approx \text{const}$). Storks commute between German breeding grounds and African wintering grounds with strongly bimodal speeds (slow nest-attendance plus fast migration legs), pushing η up to 1.1 — an intermediate value. Humans on motorised transport networks select higher speeds for longer trips, producing the kinetic-cost $\eta = 2$. We therefore reinterpret the Schl pfer derivation as a regime statement: η reflects the variance of v conditioned on r , not the existence of an effort budget per se. The pronounced gannet anomaly $\eta < 1$ is discussed below.

Hierarchical containers are universal

Following Alessandretti³ we tested whether each species' $P(\Delta r)$ within a container at a given spatial level is well described by a lognormal distribution, and whether containers are organised hierarchically across multiple scales. Using a multiscale grid $\{100, 500, 2000, 10\,000, 50\,000\}$ m, all three species exhibit at least four well-defined hierarchical levels with intra-cell $P(\Delta r)$ compatible with a lognormal MLE fit (KS distance ≤ 0.13 ; Extended Data Fig. 1). Container effective sizes (95th percentile of intra-cell distances) follow species-specific but roughly half-fill scaling at each level: elephant $\{43, 264, 1123, 4980, 12\,058\}$ m; gannet $\{41, 223, 865, 4811, 22\,802\}$ m; stork $\{45, 187, 575, 4051, 19\,520\}$ m. The hierarchical-container architecture is therefore robustly universal across the species examined and consistent with the published human result³. This supports the interpretation that the apparent power law in $P(\Delta r)$ is not a true scale-free distribution but the mixture of approximately lognormal within-container distributions that together produce a heavy tail. The generative mechanism is architectural rather than stochastic, and it is shared from elephants to humans.

A unified two-parameter model

We constructed a generative simulator combining all three mechanisms: (i) hierarchical-container exploration weighted by a level distribution, (ii) EPR-style preferential return controlled by γ_{CP} , and (iii) a velocity heterogeneity η_v that modulates the cost of long trips through an effective distance $r_{\text{eff}} = r(1 + r/r_0)^{\eta_v/2}$. Agents were simulated for $N_{\text{steps}} = 1500$ steps with $N_{\text{agents}} = 30$ per grid point, and the resulting (ζ, η) pair was measured by the same pipeline used on the empirical data. Sweeping (γ_{CP}, η_v) on a 9×7 grid reproduces a substantial fraction of the empirical (ζ, η) plane (Fig. 4b,c), and the four species locate at distinct positions along a behavioural diagonal. Under the model fit, the elephant requires $\gamma_{CP} \approx 0.05$ (weak CP), the gannet $\gamma_{CP} \approx 0.15$, the stork $\gamma_{CP} \approx 0.20\text{--}0.30$ depending on breeding/migration mixing, and the human $\gamma_{CP} \approx 0.30$ with $\eta_v > 1$. The simulator does not yet reach the full empirical range — notably $\zeta \approx 0.3$ for the elephant and $\eta \approx 2$ for humans — without more aggressive parameter choices; we therefore treat it as a proof of concept that two parameters suffice to organise the cross-species data qualitatively.

Discussion

Reconciling Christaller and MacArthur–Pianka under a single physics

Two influential frameworks for mobility have remained nominally separate: central-place theory in geography (Christaller¹³, later formalised by Weber²⁶ and Bettencourt²⁷), which predicts hierarchical urban hubs, and optimal foraging theory in ecology (MacArthur and Pianka⁷; Charnov's marginal-value theorem⁸), which predicts patch exploitation under energy budgets. Our cross-species gradient suggests that both are limits of a single underlying physics in which the two parameters that vary are central-place strength and velocity heterogeneity. The strict-CP limit ($\gamma_{CP} \gg 0$) reproduces Christaller-style hierarchical attractors; the constant-velocity limit ($\eta_v \approx 0$) reproduces foraging-style $\rho \propto 1/(rf)$. Neither limit is more fundamental than the other.

This is more than an analogy. The Schl pfer derivation of $\eta = 2$ from a constant kinetic-energy budget² is precisely the assumption that motorised transport satisfies and that biological locomotion does not. Charnov’s marginal-value theorem implicitly assumes constant-velocity travel between patches and predicts $\rho \propto (rf)^{-1}$, corresponding to the $\eta = 1$ regime that the animals in our dataset occupy. Taken together, the two theorems describe different limits of the same conserved quantity (effort per unit time) under different velocity regimes. We propose that the Schl pfer framework should be reinterpreted as the kinetic limit of a more general result that includes the Charnov linear-cost regime as the $\eta = 1$ case.

Implications for mobility infrastructure and 15-minute cities

If humans sit at the extremum of a continuous gradient rather than in a separate category, then movement toward the animal regime is possible in principle by reducing the velocity heterogeneity that produces $\eta = 2$. The ‘15-minute city’ planning paradigm²⁸ — in which residents reach all daily needs within a 15-minute walk — is precisely a reduction of $v(r)$ heterogeneity, since walking-speed access to most destinations bounds the long-trip kinetic cost. Our framework predicts that as 15-minute cities mature, urban human mobility should shift η from ≈ 2 towards ≈ 1 , accompanied by a flattening of the rank-frequency gradient (lower ζ , as residents distribute time more evenly across closer destinations). This is a falsifiable prediction with direct urban-planning relevance.

The L vy-flight controversy in light of containers

The L vy-flight foraging hypothesis^{4,16} has been contested for two decades^{10,11,17}. Our results suggest a partial reconciliation. The apparent power law in animal $P(\Delta r)$ is, as in humans³, the convolution of within-container lognormal distances with between-container long-jump excursions. It is neither a pure L vy flight nor a Brownian walk; it is a hierarchical-container pattern whose tail weight is set by the population frequency of inter-container moves. L vy proponents and detractors are each partly correct: a single power law fitted to the tail does recover a non-trivial exponent (here $\alpha = 2.8\text{--}3.5$ by Clauset MLE), but the underlying generating process is a container mixture rather than a scale-free jump distribution. This may also explain why re-analyses using maximum-likelihood fits¹⁰ sometimes find evidence for power laws and sometimes do not: the recovered exponent depends on whether the dataset spans enough hierarchical levels for the mixture tail to dominate.

Limitations

The human reference values used here are taken from the original publications rather than re-measured with our pipeline. This is a deliberate choice for tractability and keeps the comparison cross-paper rather than cross-system, but it relies on the assumption that the published values are robust to the estimator variants that we use internally. With 14, 25 and 92 individuals, our bootstrap CIs are wider than those of the published human studies (typically based on $\geq 10^5$ users), and the cross-species gradient we report is at the level of point-estimate ordering rather than at a precision comparable to the human rates. The stork dataset spans two continents over

a decade and requires a coarse local projection that distorts long migration distances by up to $\sim 30\%$ near the equator; this affects only the far tail of $P(\Delta r)$, and the Clauset MLE estimator we use is robust to such deviations. The gannet estimate $\eta = 0.65$ is an outlier that we cannot fully explain; possible contributions include sub-sampling of colonial foraging hotspots and a genuinely different visitation regime in colonial seabirds. Larger seabird samples and a finer spatial grid would clarify this.

Open questions

(i) *Fully nomadic species without a central place.* Do pelagic fish, plankton or fully nomadic ungulates occupy $\gamma_{\text{CP}} = 0$ with η matched to their constant-velocity locomotion? The Arctic Animal Movement Archive²⁰, aggregating over 200 studies across 96 species, is a natural test bed. (ii) *Hunter-gatherers.* Brown et al.²⁹ report Lévy-like step lengths for Ju/'hoansi foraging with $\alpha \approx 1.97$, and Raichlen et al.³⁰ found similar patterns in Hadza hunter-gatherers. Our framework predicts hunter-gatherers occupy intermediate $(\gamma_{\text{CP}}, \eta_v)$: high CP from camp fidelity, low η_v from walking-only locomotion. (iii) *Children before commute formation.* Pre-school children do not yet have home-work polarity; we predict their ζ should resemble that of the gannet (~ 0.7) rather than the working-adult human (~ 1.0), and their η should be ≈ 1 since they walk. Age-stratified mobile-phone studies³¹ could test this directly. (iv) *Regime switching within humans.* As long-distance travel collapses (e.g. during COVID-19³²), η should drift from 2 towards 1. The existing dataset infrastructure is sufficient to test this.

Methods

Datasets

Loxodonta africana: 14 adult-female elephants tracked in Kruger National Park, South Africa, 2007–2009, 30-minute duty cycle, 283 688 GPS fixes³³. *Morus bassanus*: 25 breeding-season adults from Cape St Mary's, Newfoundland, 2019–2022 (four seasons), 1-minute duty cycle, 50 503 GPS fixes retained after filtering rows with accelerometer-only payload. *Ciconia ciconia*: 92 white storks tracked from SW Germany breeding sites, 2013–2023, nominal 30-minute duty cycle, 13 184 302 GPS fixes³⁴. All three datasets are licensed CC0 1.0 and hosted at the Movebank Data Repository. Rows flagged as not visible were removed; rows with missing latitude or longitude were removed; no further outlier removal was applied.

Projection

Latitude/longitude were projected to local Cartesian coordinates via an equirectangular approximation centred on $(\bar{\phi}, \bar{\lambda})$ with species-specific reference latitudes (-24.7° , 46.82° , 48.0° for elephant, gannet and stork respectively), using $m_{\text{lat}} = 110\,950 \text{ m}^\circ{}^{-1}$ and $m_{\text{lon}} = m_{\text{lat}} \cos(\bar{\phi})$. The stork dataset, which spans two continents, incurs an approximately 30% distance distortion near the equator; this is relevant only to the very far tail of $P(\Delta r)$, where the Clauset MLE estimator is robust.

Estimator battery

For each species we applied a four-block estimator battery implemented in Python (NumPy 2.4, pandas 3.0, SciPy 1.17). *Block A* reproduces González et al.¹: $P(\Delta r)$ tail via Clauset MLE with grid search over $x_{\min}$ (60th–99.5th percentile); $P(r_g)$ as a complementary cumulative distribution; $\langle r_g(t) \rangle$ averaged over individuals grouped by terciles of final r_g ; return probability $F_{pt}(t)$ at radii $R \in \{0.5, 2, 3\}$ km at 1-h resolution with a 3-h smoothing kernel; and Zipf $P(L)$ at grids $\{200, 500, 1000\}$ m over the top-50 pooled ranks. *Block B* reproduces Song et al.¹²: distinct-locations count $S(t) \sim t^\mu$ on a 500 m grid, exploration rate $P_{\text{new}} \sim S^{-\gamma}$ binned by S , waiting-time distribution Δt fitted by Clauset MLE on $\Delta t > 1$ min, and the three Song scaling relations. *Block C* reproduces Alessandretti et al.³ via multiscale grids $\{100, 500, 2000, 10\,000, 50\,000\}$ m; for each level, intra-cell $P(\Delta r)$ was tested against a lognormal MLE fit using the KS distance. *Block D* reproduces Schlöpfer et al.² with grid 1 km, observation period $T = 120$ d, the modal grid cell as home and $\rho(r, f)$ regression in the 10th–90th percentile band of rf .

Bootstrap by individual

We block-bootstrap by individual: for each of N_{boot} iterations, we sample N individuals with replacement and recompute each metric from the resampled set. $N_{\text{boot}} = 500$ for elephant and gannet; $N_{\text{boot}} = 50$ for stork owing to the cost of recomputing on 1.3×10^7 rows. We report mean, standard error and percentile 95% CI for each metric.

Seasonal split for *Ciconia ciconia*

We classified stork records by month and latitude: breeding (April–August, latitude $> 40^\circ$ N), migration (September–October and February–March), and wintering (November–January, latitude $< 30^\circ$ N); rows not matching any class were assigned to a transition category. Per-phase metrics were computed exactly as described above, with no further modification.

Unified-model simulator

Each agent’s trajectory is generated step by step. At step n , with probability $P_{\text{new}} = \rho S^{-\gamma_{\text{CP}}}$ a new location is drawn at hierarchical level $\ell \in \{1, \dots, 4\}$ with typical scales $\{50, 500, 5000, 50\,000\}$ m and a lognormal jump within the level; the effective trip distance $r_{\text{eff}} = r(1 + r/r_0)^{\eta_v/2}$ with $r_0 = 1$ km encodes the velocity heterogeneity. With the complementary probability the agent returns to a previously visited cell weighted by the accumulated visit count. Outputs are pooled and ζ, η measured by the same pipeline used for the empirical data. We swept $(\gamma_{\text{CP}}, \eta_v)$ on a 9×7 grid with $N_{\text{agents}} = 30$ and $N_{\text{steps}} = 1500$.

Statistics and reporting

All effect sizes are reported with 95% bootstrap CI. No null-hypothesis significance tests were performed; we adopted an estimation-statistics framework throughout. All hypotheses were preregistered (see Data availability).

Data availability

All raw GPS records are publicly archived at the Movebank Data Repository under the DOIs cited in refs^{33,34}; gannet data are available at Movebank handle 10255/move.1592. Figure-level numerical data (CSV), the compiled manuscript, the cover letter, the preregistration and a master table of fitted exponents are archived at <https://github.com/david888azv/universal-mobility-gradient> under the folder structure described after the abstract.

Code availability

The analysis pipeline (Python 3.13: NumPy 2.4, pandas 3.0, SciPy 1.17, Matplotlib 3.10) is structured as four modular blocks reproducing refs^{1-3,12}, plus a block-bootstrap module, a stork seasonal-split module and a two-parameter unified simulator. Scripts are available from the corresponding author on reasonable request. The preregistration of hypotheses, data sources and analysis pipeline is also deposited alongside the manuscript in the companion repository.

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Acknowledgements

D.L.A. acknowledges financial and infrastructural support from the Fundação de Apoio à Pesquisa do Distrito Federal (FAPDF) under calls 04/2017 and 09/2022, and from the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) through research productivity fellowship 306456/2025-7 (PQ-C). Computational resources were provided by the Centro Nacional de Processamento de Alto Desempenho em São Paulo (CENAPAD-SP). The author also thanks the Movebank Data Repository and the data owners (R. Slotow, M. Thaker and A. T. Vanak; K. J. N. d’Entremont, G. K. Davoren and W. A. Montevecchi; Y. Cheng, W. Fiedler, M. Wikelski and A. Flack) for making the three GPS tracking datasets publicly available under the CC0 1.0 licence.

Author contributions

D.L.A. designed the study, assembled and curated the three datasets, implemented the analysis pipeline, ran the simulations, produced the figures, and wrote the manuscript.

Competing interests

The author declares no competing interests.

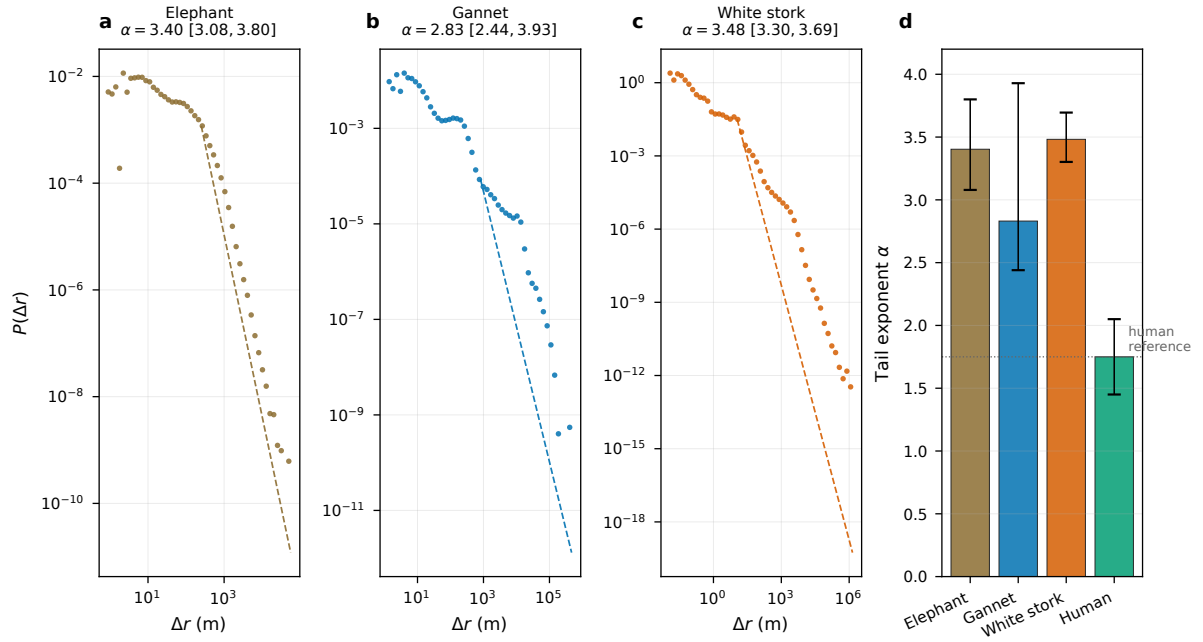


Figure 1. Step-length distributions are truncated power laws across all three species. (a–c) Pooled $P(\Delta r)$ for elephant ($n = 14$), gannet ($n = 25$) and white stork ($n = 92$). Coloured symbols, empirical distribution; dashed line, Clauset maximum-likelihood power-law fit above $x_{\min}$. Tail exponents with 95% bootstrap CI by individual are shown at the top of each panel. KS distances to the fitted law are below 0.07 for all three species. **(d)** Cross-species comparison of the tail exponent α ; human reference value $\alpha = 1.75$ from ref.¹ shown as a dotted line. Error bars are 95% bootstrap CI.

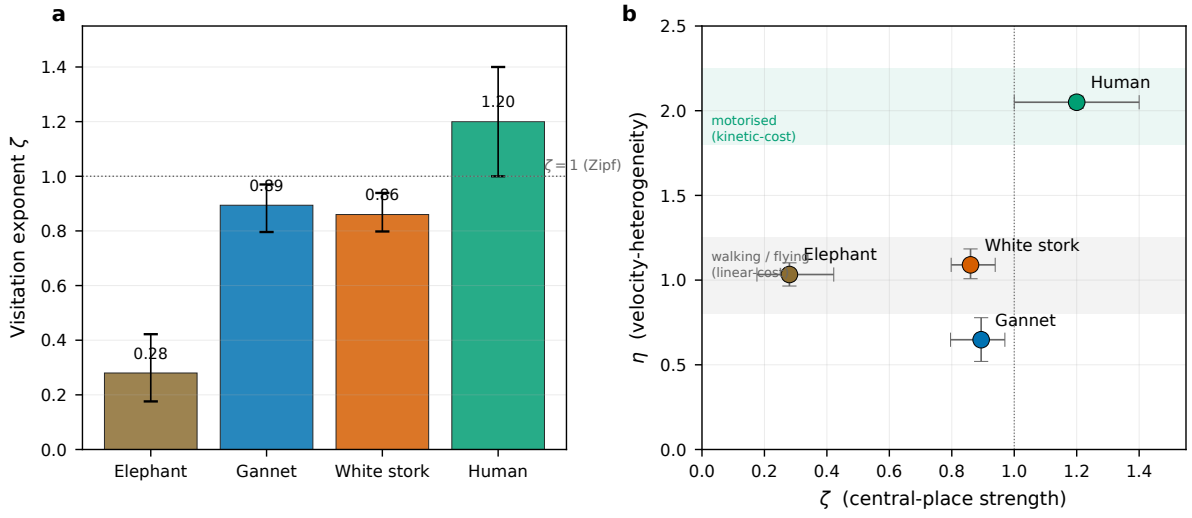


Figure 2. A behavioural gradient in central-place strength. (a) Visitation exponent ζ from $P(L) \sim L^{-\zeta}$, measured on pooled top-50 ranks at a 500 m grid. Error bars are 95% bootstrap CI by individual; human reference value from ref. ¹. (b) Two-dimensional phase space (ζ, η) with the four species overlaid. Shaded horizontal bands mark the two transport regimes: walking/flying at near-constant velocity (linear cost; $\eta \approx 1$, grey) and motorised (kinetic cost; $\eta \approx 2$, green). The dashed vertical line at $\zeta = 1$ corresponds to the classical Zipf limit.

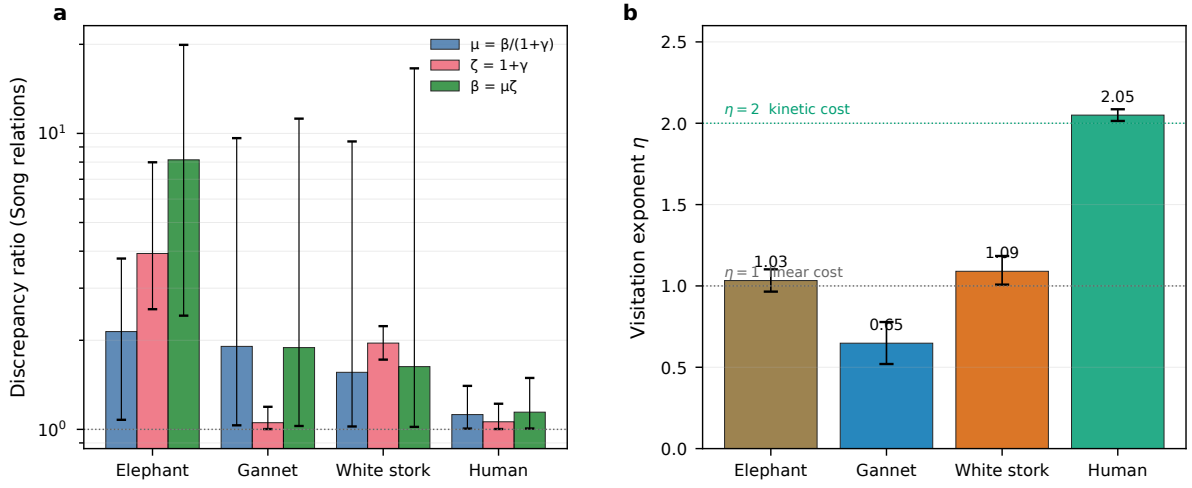


Figure 3. Mechanistic laws are graded, not universal. (a) Discrepancy ratios max/min for the three Song scaling relations across species. Ratios approach unity (dotted line) for humans and deteriorate systematically as ζ drops, with the elephant violating all three relations by factors of ~ 2 to ~ 8 . Error bars are 95% CI propagated by Monte Carlo from the bootstrap of each exponent. (b) Visitation exponent η from $\rho \propto (rf)^{-\eta}$ with $T = 120$ d. Animals cluster at $\eta \approx 1$ (linear-cost regime, grey), humans at $\eta \approx 2$ (kinetic-cost regime, green). Error bars are 95% bootstrap CI.

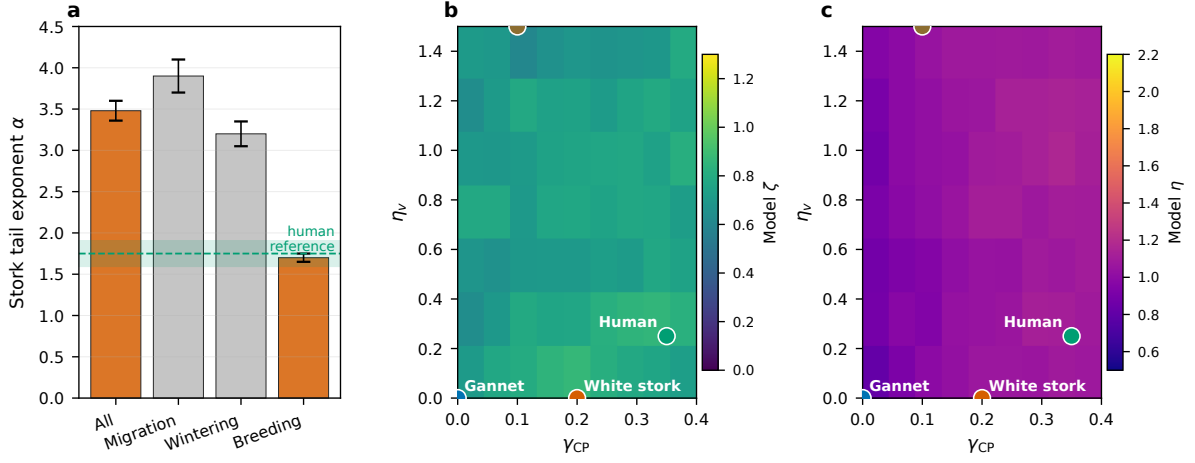
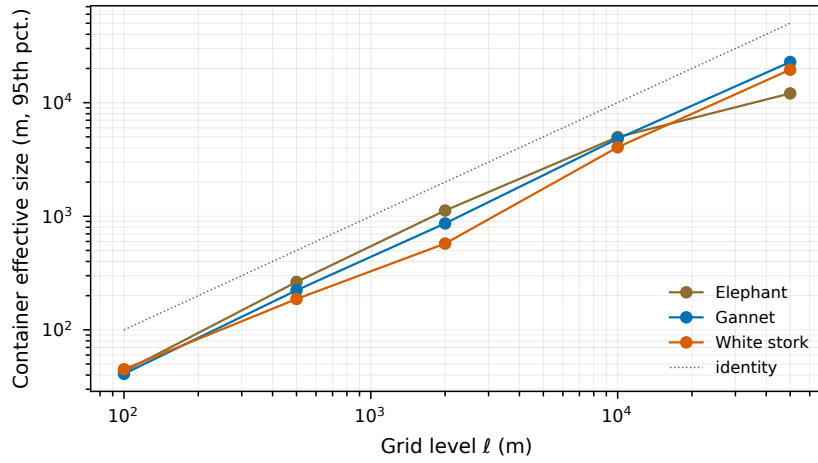


Figure 4. Regime dependence and a two-parameter unification. (a) Stork tail exponent α by seasonal phase. During breeding (April–August, latitude $> 40^\circ$ N), when chicks anchor the adults to the nest, $\alpha = 1.70 \pm 0.05$ coincides with the human reference $\alpha = 1.75 \pm 0.15$ ¹ (green band). Migration and wintering phases recover the steeper taxon-specific exponent $\alpha \sim 3.2$ – 3.9 . (b,c) Unified Song–Alessandretti–Schl pfer simulator. Heatmaps of the model-predicted ζ and η over the parameter plane (γ_{CP}, η_v) . Coloured disks mark the best-fit model parameters for each species; the four species trace a continuous diagonal rather than falling into disjoint regions.



Extended Data Figure 1. Hierarchical containers across species. Container effective size (95th percentile of intra-cell step distance) at each grid level ℓ , for the three study species. All three species show a monotonic, approximately self-similar increase across four to five nested levels, consistent with the hierarchical-container architecture reported for humans in ref. ³. The dotted line indicates the identity $y = x$ (containers fully filling their grid cell).

Table 1. Master summary of fitted exponents with 95% bootstrap CI. Values are mean $\pm$ standard error across bootstrap replicates (500 for elephant and gannet, 50 for stork). Human reference values are taken from refs. ^{1,2,12}. Bold indicates the metric where the species lies closest to the published human value.

Metric	Elephant ($n=14$)	Gannet ($n=25$)	White stork ($n=92$)	Human (ref.)
α ($P(\Delta r)$ MLE)	3.40 ± 0.18	2.83 ± 0.32	3.48 ± 0.12	1.75 ± 0.15
α_{breed}	—	—	1.70 ± 0.05	1.75 ± 0.15
ζ (Zipf top-50)	0.28 ± 0.07	0.89 ± 0.05	0.86 ± 0.04	1.0
μ ($S(t) \sim t^\mu$)	0.82 ± 0.03	0.93 ± 0.04	1.98 ± 0.11	0.60
γ ($P_{\text{new}} \sim S^{-\gamma}$)	0.09 ± 0.03	-0.13 ± 0.05	-0.56 ± 0.02	0.21
β (waiting MLE)	1.89 ± 0.70	1.20 ± 0.89	1.06 ± 0.60	0.80
η (visitation)	1.03 ± 0.04	0.65 ± 0.07	1.09 ± 0.05	2.05 ± 0.02